

A report on

Human Activity Recognition (HAR) Using Integrated Spatio-Temporal Analysis.

By

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To

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Prepared in fulfillment of the course

BITS F422T

Birla Institute of Technology & Science Pilani

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Introduction

This report contains a detailed explanation of the methods employed to do teacher analysis on CCTV footage. In the first part of this report, I have discussed the conceptual and structural detail of the model employed. In this part, I have discussed the rule-based system's working and then a brief discussion on what exactly I have done on the data and what are the results I got to form it. In the second part of the report, I have discussed the space-time method in which we have used convolution in addition to the LSTM model for analysis of the video (convolution neural network for analysing the spatial part of data and LSTM for analysing the temporal part of data). In the Third part of the report, we have discussed the result obtained from using these methods in which we have first discussed the general results like finding a region of interest and a heat map of the video. And after that, I discussed specifically what is the result I got from each method: - rule-based (got landmark of person tracked and their association with different activities) and space-time (got feature of interest and activity class name). In the end, we have discussed the overall result that we got from the analysis and what are the shortcoming of both methods and how to overcome them. Finally, I discussed what to do next and how this project will benefit the school.

In recent years, Human Activity Recognition (HAR) has emerged as a critical field at the intersection of computer vision, machine learning, and real-time systems. The ability to automatically recognize and analyze human activities has profound implications across various domains, from security and surveillance to healthcare and retail analytics. This research focuses on developing an integrated spatio-temporal framework for robust human activity recognition, specifically addressing the challenges of real-time processing and multi-camera environments.

The exponential growth in surveillance infrastructure, particularly in urban environments, has created both opportunities and challenges. While we now have access to unprecedented amounts of visual data, processing this information effectively and extracting meaningful insights remains a significant challenge. Traditional approaches to activity recognition often struggle with real-world complexities such as varying lighting conditions, occlusions, and the need for real-time processing.

Our research addresses these challenges through a comprehensive approach that integrates advanced hardware infrastructure with sophisticated software solutions. By leveraging state-of-the-art deep learning techniques and modern

cloud computing capabilities, we aim to create a system that can perform reliable activity recognition while maintaining real-time performance.

Key Challenges in Current HAR Systems:

- Real-time Processing: Existing systems often struggle to balance accuracy with processing speed, particularly when handling multiple video streams simultaneously.
- Multi-camera Integration: Coordinating and synchronizing data from multiple cameras while maintaining temporal consistency presents significant technical challenges.
- Scalability: Many current solutions fail to scale effectively when deployed in large-scale environments with numerous cameras and users.
- Accuracy in Complex Scenarios: Performance degradation in challenging conditions such as poor lighting, occlusions, and crowded scenes remains a persistent issue.

Our research proposes a novel framework that addresses these challenges through a carefully designed architecture combining edge computing with cloud-based processing. The system utilizes a network of high-resolution CCTV cameras (5MP at 20FPS) connected to local processing units, which perform initial data preprocessing and feature extraction. This processed data is then streamed to cloud infrastructure using optimized RTSP protocols, enabling sophisticated analysis while maintaining real-time performance.

The significance of this research lies in its practical applicability and potential impact across various sectors. For instance, in security applications, the ability to accurately recognize and track activities in real-time can significantly enhance surveillance capabilities. In retail environments, understanding customer behavior patterns can provide valuable insights for business optimization. Healthcare facilities can benefit from automated monitoring systems that can detect and alert staff to potentially dangerous situations.

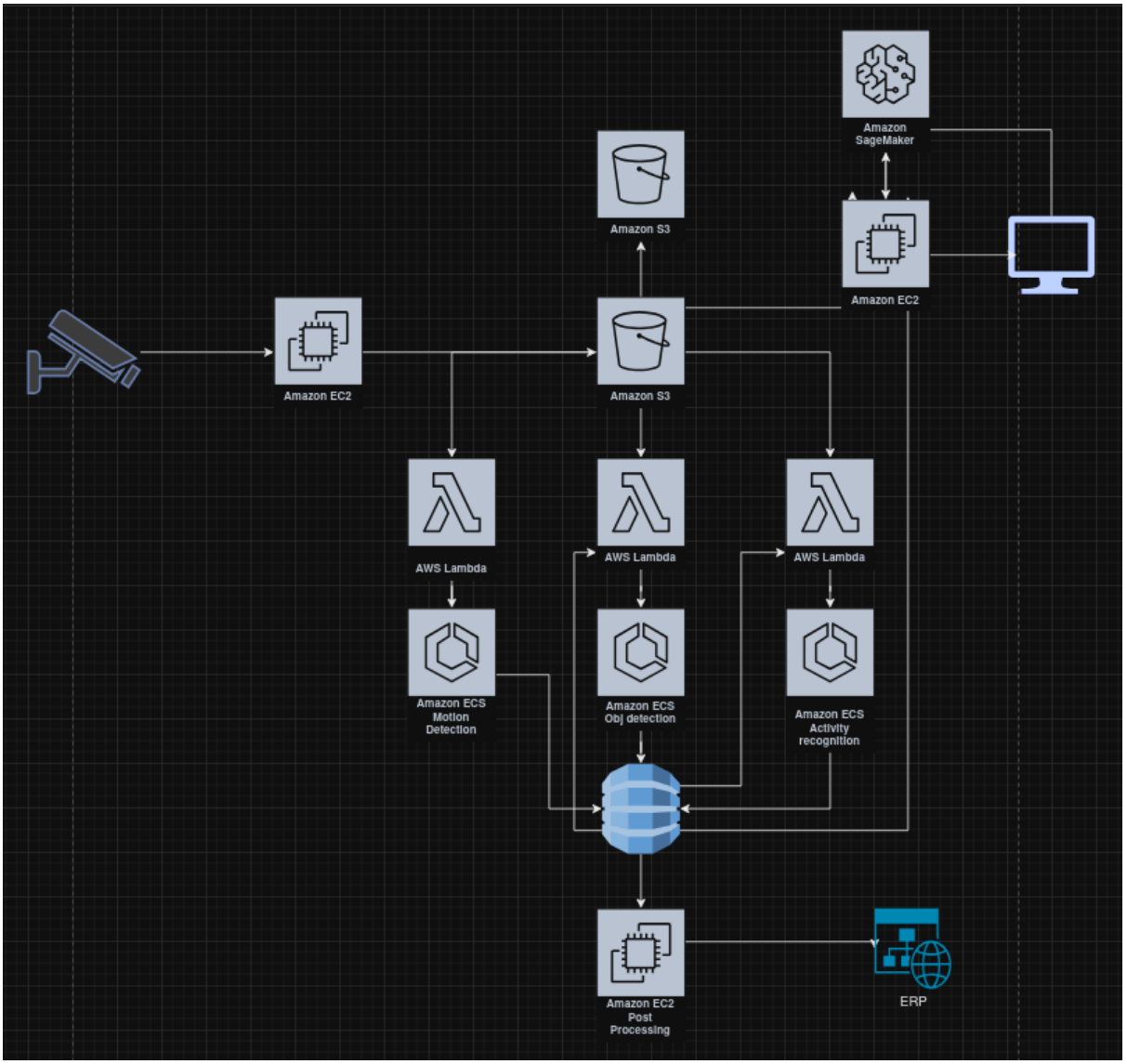
Our approach distinguishes itself through several innovative aspects:

1. Integrated Edge-Cloud Architecture: By intelligently distributing processing tasks between edge devices and cloud infrastructure, we achieve optimal performance while managing computational resources effectively.
2. Advanced Temporal Analysis: The system incorporates sophisticated temporal modeling techniques to maintain consistency across video frames and multiple camera views.
3. Scalable Design: Our architecture is designed to scale seamlessly, accommodating additional cameras and processing requirements without significant performance degradation.
4. Robust Error Handling: Comprehensive error detection and recovery mechanisms ensure system reliability in real-world deployments.

The remainder of this thesis is organized to provide a detailed examination of our research methodology, implementation, and results. We begin with a comprehensive overview of the system architecture, followed by detailed discussions of our model development process and deployment strategies. The results section presents both quantitative and qualitative analyses of system performance, concluding with insights into future research directions and potential applications.

Through this research, we aim to contribute to the advancement of HAR systems by demonstrating a practical, scalable approach that addresses real-world challenges while maintaining high accuracy and performance standards. Our findings not only provide valuable insights for researchers in the field but also offer practical solutions for organizations looking to implement advanced activity recognition systems.

System Architecture



The foundation of our Human Activity Recognition system lies in its carefully designed architecture that seamlessly integrates edge hardware with cloud infrastructure. This chapter details the complete system architecture, emphasizing how different components work together to achieve efficient real-time processing and scalable activity recognition.

Hardware Infrastructure

Our system employs a distributed network of surveillance cameras strategically positioned to provide comprehensive coverage. The primary surveillance infrastructure consists of 16 HIKVISION cameras, divided into two distinct configurations. The first set comprises eight 5-megapixel cameras operating at 20 frames per second, providing high-resolution coverage for critical areas. The second set includes eight 3.45-megapixel cameras, also operating at 20 frames per second, deployed in areas requiring standard monitoring. This dual-configuration approach optimizes both storage requirements and processing demands while maintaining effective coverage.

For edge computing, we utilize two dedicated processing units, each equipped with Intel i5 7th generation processors and 8GB RAM. These units are built on Zebronics motherboards and include 128GB storage for temporary data handling. The edge units serve as the first layer of processing, handling initial video capture and preprocessing before transmission to cloud infrastructure. This configuration was chosen to balance cost-effectiveness with processing capabilities, providing sufficient power for real-time video handling while maintaining system reliability.

Network connectivity forms a crucial component of our infrastructure. The system operates on a dual-ISP configuration, utilizing Airtel Fiber Optic for primary connectivity and a local ISP as backup. This redundant network setup ensures continuous operation and provides the necessary bandwidth for real-time data transmission. The network topology is designed to minimize latency while maintaining data integrity, with dedicated channels for video streaming and system management.

Data Pipeline

The data pipeline represents the core of our system's operational workflow. Video streams from all cameras are captured using the Real-Time Streaming Protocol (RTSP), chosen for its reliability and low-latency characteristics. Our implementation includes custom buffer management to handle network fluctuations and ensure smooth video delivery.

A key innovation in our pipeline is the multi-camera synchronization system. Rather than treating each camera feed independently, we've developed a synchronized processing approach that maintains temporal alignment across all feeds. This synchronization is crucial for accurate activity recognition across multiple viewpoints and enables coherent multi-camera tracking.

The real-time processing workflow follows a staged approach:

1. Initial frame capture and buffering at the edge units
2. Preliminary motion detection and frame selection
3. Batch processing of selected frames for efficiency
4. Synchronized transmission to cloud infrastructure

Cloud Integration

The cloud component of our architecture leverages Amazon Web Services (AWS) to create a scalable and reliable processing environment. The system employs a serverless architecture pattern, minimizing operational overhead while maximizing processing efficiency.

Our AWS implementation follows a sophisticated event-driven architecture:

- EC2 instances serve as the primary ingestion points for video streams from edge units
- S3 storage is strategically utilized across different tiers (Standard, Glacier, and Deep Archive) based on data access patterns
- Lambda functions trigger processing workflows based on data arrival
- AWS Fargate handles compute-intensive tasks like motion detection and activity recognition
- DynamoDB stores metadata and processing results, enabling quick access and analysis

The storage strategy employs a tiered approach, with recent data kept in standard S3 storage for quick access, while historical data is automatically moved to more cost-effective storage tiers like S3 Glacier and S3 Deep Archive. This approach optimizes storage costs while maintaining data accessibility.

For compute resources, we utilize AWS Fargate for containerized workloads, allowing automatic scaling based on processing demands. The system dynamically adjusts computing resources based on:

- Current processing load
- Number of active camera feeds
- Complexity of detected activities
- Time-of-day patterns in activity levels

Scalability considerations have been fundamental to our architectural decisions. The system is designed to handle:

- Horizontal scaling of camera inputs
- Variable processing loads
- Burst activity periods
- Geographic distribution of processing

The architecture's modularity allows for easy addition of new cameras or processing nodes without system redesign. Load balancing is handled automatically through AWS services, ensuring optimal resource utilization and consistent performance.

Methods used for Analysis

In this project I have tried to analyse video by three different methods out of which one is rule based method and another 2 is space time waste method I hear I have attempted to briefly explain you the details of all three methods and how I have applied them the code for all of three method is provided with them

Rule-Based Methods (Code)

A rule-based approach determines an ongoing event by modelling the activity using a set of rules or attributes that describe the event. Each activity is viewed as a set of basic rules/attributes so that descriptive models can be built to detect human activity.

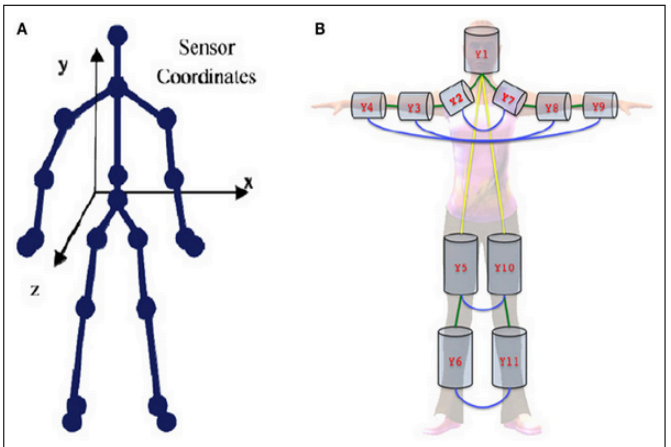


Figure 1. Human body representations. (A) 2D skeleton model (Theodorakopoulos et al., 2014) and **(B)** 3D pictorial structure representation (Belagiannis et al., 2014).

Action detection for complex multithematic scenes was proposed by Morariu and Davis (2011). Each subject must follow a specific set of rules when performing actions. The detection process was run on a video of a basketball game, where players were first detected and tracked, generating a set of trajectories that were used to create a series of spatio-temporal events. Based on first-order logic and probabilistic approaches such as Markov networks, the authors were able to infer which event occurred. Figure 1 summarises a method using primitive rules to detect human behaviour. Liu et al. (2011a) addressed the problem of recognizing behaviour based on a set of descriptive and discriminatory attributes. Each attribute is mapped to a feature that describes the spatiotemporal nature of the activity. These attributes were treated as latent variables to capture the level of importance of each attribute for each action in our potential SVM approach.

How did I used it in this project?

First of all I did some pre-processing on the video. I clipped the video in 20 second short videos so for this purpose of analysis I used for 48 minute and 22nd long video which was divided in short 142 short videos of 20 second each

Then I applied a pose detection algorithm by Media Pipeline from Google to the video and got the X ,Y , Z coordinate and visibility of different points of the tracked object. (LInk to full sheet)

MediaPipe Pose is only used for initial analysis of the problem. Return each landmark with its ID, where each ID corresponds to a different body part. (Fig. 4.3). From this we can develop a rule-based system for activity analysis, but this is very tedious and not scalable. This rule-based learning approach was researched in 2015 and works wonders in many situations, but it relies heavily on accurate body point detection, which doesn't work in our case.

Graph below, we can see that the teacher's speed is changing rapidly. This is the result of analysing only short clips. We analysed only a clip of her six-second video in which the teacher was already moving. After a while you will find that the teacher may not move and this graph collapses to zero. This shows the teacher's involvement in the class. B. When the teacher walks around the class and interacts with everyone. This also lets you know if your kids are walking around the classroom during class, socialising and playing games during the fun times. This can be used to determine cross-class involvement.



Figure 2 :- Result for Rule based System

Space-Time Methods (Code)

Spatiotemporal approaches focus on recognizing motion based on spatiotemporal characteristics or trajectory matching. They consider activity in a three-dimensional space-time volume composed of temporal connections in two-dimensional space. Activity is represented as a series of spatiotemporal features or trajectories extracted from footage. Figure 3 shows an example of a spatiotemporal approach based on dense trajectories and motion descriptors

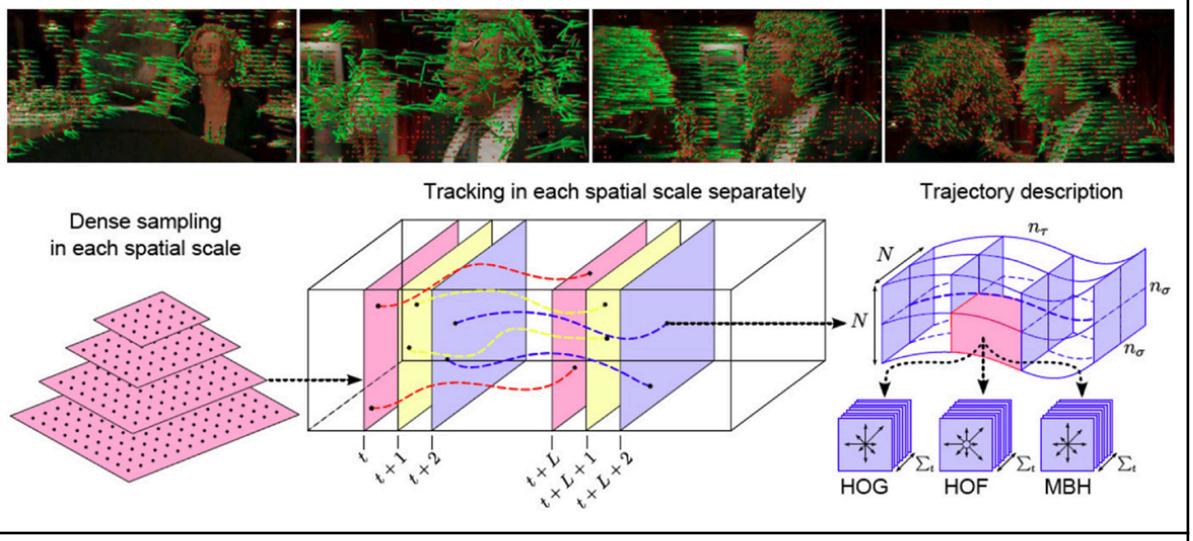


Figure 3 Visual representation of Conv-LSTM

Before getting into the structure Conv-LSTM of it would be wise to understand the convolution neural network separately and long-term short-term memory network separately so the following section will give you a brief introduction about convolution your network and LSTM

Long-Short Term Memory Network

This is really help full video on this topic for understanding finer mathematical details of this network

LSTM networks are specifically designed to work with sequence data because they consider all previous inputs when generating the output. An LSTM is actually a type of neural network called a

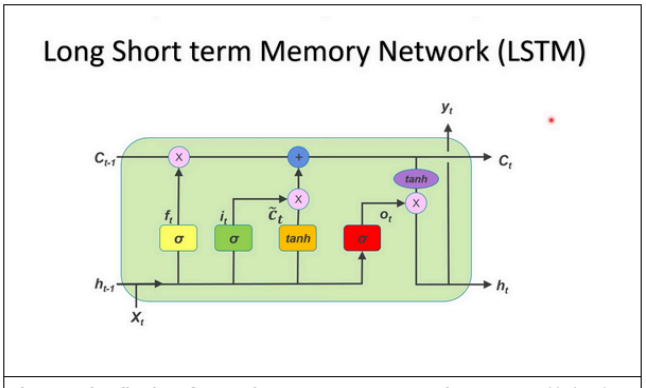


Figure 4. Visualisation of Long-Short Term Memory Network Many to one kind variant which is used in this project

recurrent neural network, but RNNs are known to be inefficient at handling long-term dependencies in input sequences due to a problem called the gradient vanishing problem.
The LSTM is designed to overcome gradient vanishing so that the LSTM cells can remember the context of long input sequences.

Convolutional Neural Network

A Convolutional Neural Network (CNN or ConvNet) is a type of deep neural network specifically designed to work with image data, and excels at analysing images and making predictions based on them.
It works with a kernel (called a filter) that scans an image and creates a feature map (which indicates whether a certain feature is at a certain location in the image). We use merge operations to increase the number of feature maps and reduce the size of the maps without losing important information in the network.

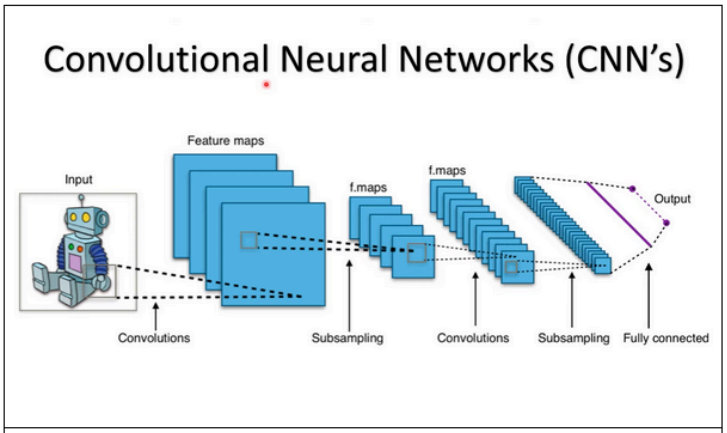


Figure 4. Visualisation of Convolutional Neural Network

Each layer of ConvNet learns increasingly complex features. That is, for example, the first layer can learn how to detect edges and corners, and the last layer can learn how to recognize people in different poses.
Now let's return to our discussion of different approaches to video classification.
For this project I have used to models which employee the same principle of convolution neural network and capturing the temporal part by an LSTM I have given details about them briefly in the following part: -

Conv-LSTM (Convolution Long-Short Term Memory Network)

Conceptual Understanding

Long-Short Term Memory Network LSTM achieve the great success in analysing and processing sequential data yielding in state of art results in recognition of sequential data like text audio signal processing etc whereas are CNN or convolution neural network has achieved a great success in recognition of special features in images etc now

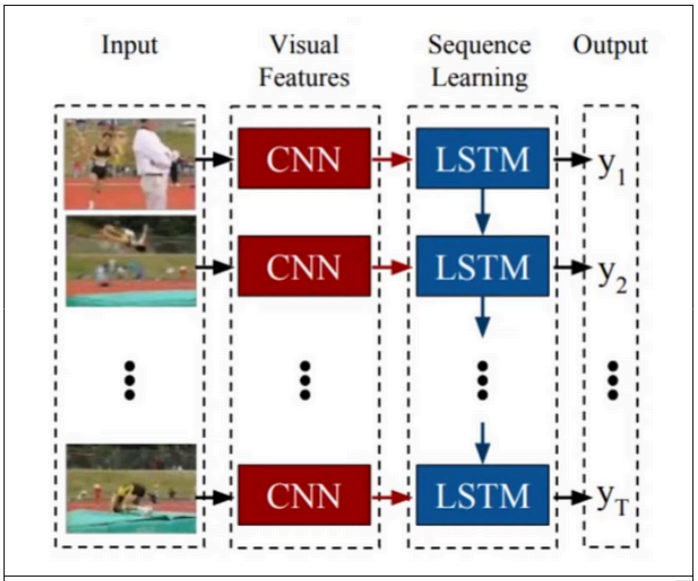
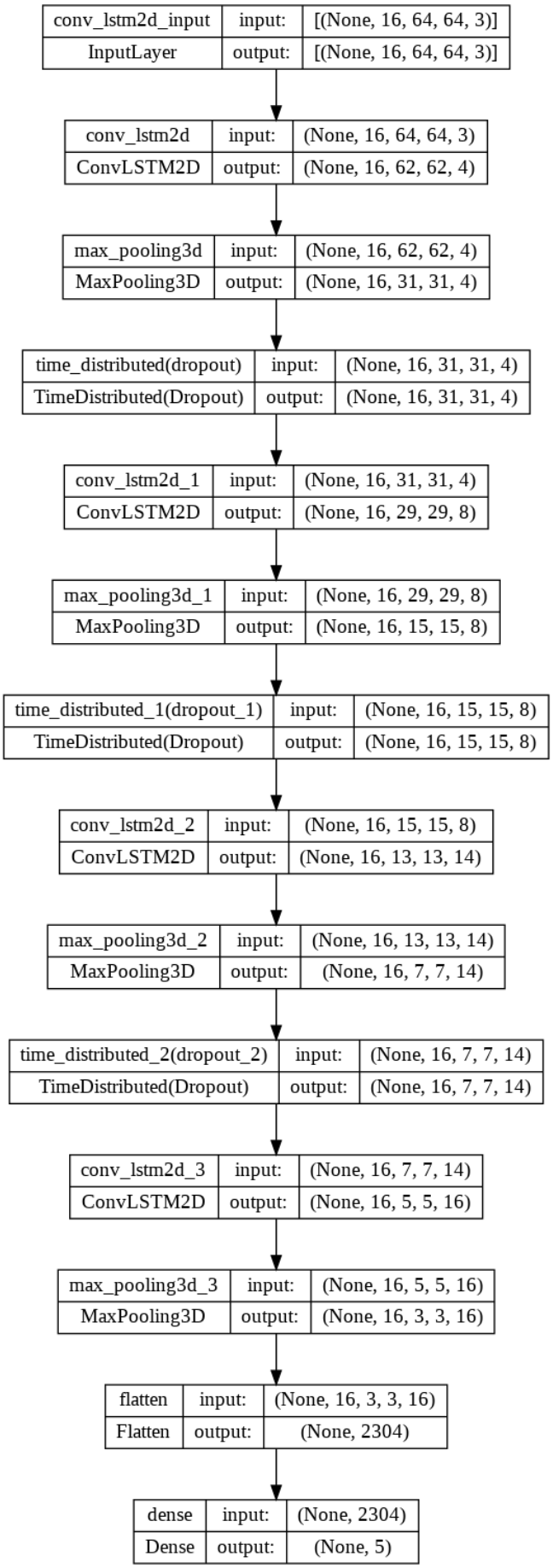


Figure 5. Visual Representation of the Conv-LSTM

In this project, we use Convolutional Neural Network (CNN) and Deep Network LSTM (D-LSTM) to process video data to perform motion recognition. First, a deep feature is extracted every 6th frame of the video to help reduce redundancy and complexity. We then use a DB-LSTM network to explore sequential information among the frame's features. Here, multiple layers are added together in both the forward and backward passes of the DB-LSTM to increase the depth. The proposed method can train long sequences and process long videos by analysing features over a certain time interval.

in the temporal part this method



Understanding structure and implementation

For this part I highly suggest to check out the [code]

On the Left hand side you can see the exact structure of the neural network and in the bottom you can see number of the parameter each layer and the total number of parameters in the network

```
Model: "sequential"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 conv_lstm2d (ConvLSTM2D)    (None, 16, 62, 62, 4)     1024

 max_pooling3d (MaxPooling3D  (None, 16, 31, 31, 4)    0
 )

 time_distributed (TimeDistr  (None, 16, 31, 31, 4)    0
 ibuted)

 conv_lstm2d_1 (ConvLSTM2D)  (None, 16, 29, 29, 8)     3488

 max_pooling3d_1 (MaxPooling  (None, 16, 15, 15, 8)    0
 3D)

 time_distributed_1 (TimeDis  (None, 16, 15, 15, 8)    0
 tributed)

 conv_lstm2d_2 (ConvLSTM2D)  (None, 16, 13, 13, 14)    11144

 max_pooling3d_2 (MaxPooling  (None, 16, 7, 7, 14)     0
 3D)

 time_distributed_2 (TimeDis  (None, 16, 7, 7, 14)     0
 tributed)

 conv_lstm2d_3 (ConvLSTM2D)  (None, 16, 5, 5, 16)      17344

 max_pooling3d_3 (MaxPooling  (None, 16, 3, 3, 16)     0
 3D)

 flatten (Flatten)           (None, 2304)              0

 dense (Dense)               (None, 5)                 11525

=================================================================
Total params: 44,525
Trainable params: 44,525
Non-trainable params: 0
_________________________________________________________________
```

In total the network have 44,525 trainable parameters and about 7 hyperparameter (which is substantially better than pure 3D convolution network whose parameters reaches can easily reach over a million)

For which loss is categorical_crossentropy, optimizer is Adam and metrics of evaluation is accuracy.

LRCN (LONG-TERM RECURRENT CONVOLUTIONAL NETWORK)

Conceptual Understanding

LRCN approach by combining convolutional and LSTM layers in one model. Another similar approach is to use separately trained CNN and LSTM models. A CNN model can be used to extract spatial features from video frames, for which a pre-trained model that can be fine-tuned can be used to solve the problem. The LSTM model can then use the features extracted from the CNN to predict actions performed on the video.

However, here we are implementing a different approach called Long Term Recurrent Convolutional Network (LRCN) which combines CNN and LSTM layers into one model. Convolutional layers are used to extract spatial features from frames, which are passed to LSTM layers at each time step to model temporal sequences. Thus, the network directly learns spatiotemporal features during end-to-end training to create reliable models.

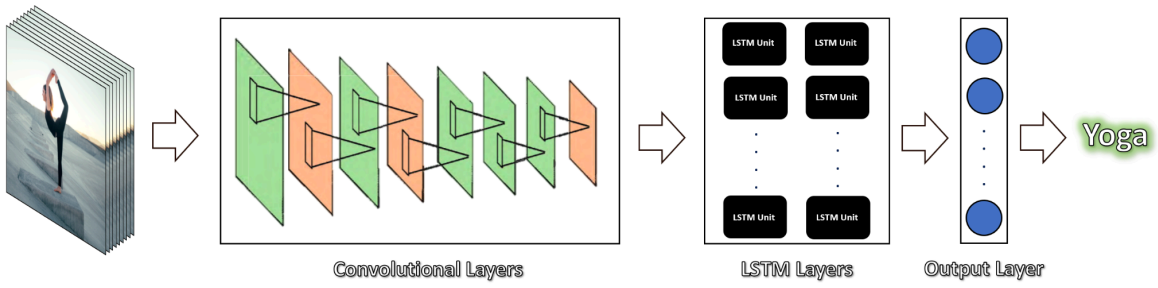
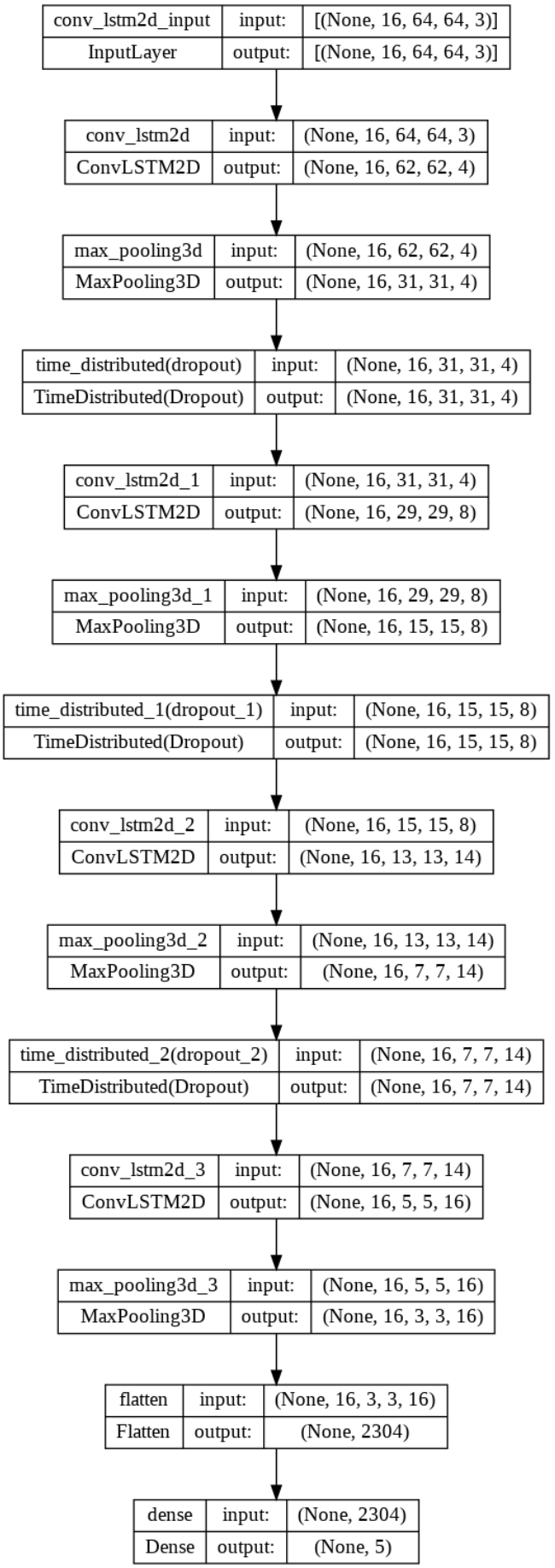


Figure 8. Visual Representation of LRCN (LONG-TERM RECURRENT CONVOLUTIONAL NETWORK)

In This Project Long-Term Recurrent Convolutional Network (LRCN) that combines a deep-layer visual feature extractor (e.g. CNN) with a model capable of learning how to recognize and synthesise temporal dynamics for tasks containing sequential data to a Long Recurring model capable of learning temporal nature of the data.



Understanding structure and implementation

For this part I highly suggest to check out the code

On the Left hand side you can see the exact structure of the neural network and in the bottom you can see number of the parameter each layer and the total number of parameters in the network

```
Model: "sequential_1"
_________________________________________________________________
 Layer (type)                 Output Shape              Param #
=================================================================
 time_distributed_3 (TimeDis  (None, 16, 64, 64, 16)   448
 tributed)

 time_distributed_4 (TimeDis  (None, 16, 16, 16, 16)   0
 tributed)

 time_distributed_5 (TimeDis  (None, 16, 16, 16, 16)   0
 tributed)

 time_distributed_6 (TimeDis  (None, 16, 16, 16, 32)   4640
 tributed)

 time_distributed_7 (TimeDis  (None, 16, 4, 4, 32)     0
 tributed)

 time_distributed_8 (TimeDis  (None, 16, 4, 4, 32)     0
 tributed)

 time_distributed_9 (TimeDis  (None, 16, 4, 4, 64)     18496
 tributed)

 time_distributed_10 (TimeDi  (None, 16, 2, 2, 64)     0
 stributed)

 time_distributed_11 (TimeDi  (None, 16, 2, 2, 64)     0
 stributed)

 time_distributed_12 (TimeDi  (None, 16, 2, 2, 64)     36928
 stributed)

 time_distributed_13 (TimeDi  (None, 16, 1, 1, 64)     0
 stributed)

 time_distributed_14 (TimeDi  (None, 16, 1, 1, 64)     0
 stributed)

 time_distributed_15 (TimeDi  (None, 16, 64)           0
 stributed)

 lstm (LSTM)                 (None, 32)                12416

 dense_1 (Dense)             (None, 5)                 165

=================================================================
Total params: 73,093
Trainable params: 73,093
Non-trainable params: 0
_________________________________________________________________
```

As you can see this is much bigger (almost double) of Conv-LSTM with about 73,093 parameters.

Results

Finding region of interest

From the above analysis, the concept of mean motion for the video can also be extracted and used in the region of interest. This is an interesting visual representation of the classroom movement scene.



Fig 10:- Region of interest in the class (Region with highest kinetic motion)

Thanks to this, it is possible to determine which areas in the classroom have the most motor movements, which can be used to analyze the movements of both teachers and students in the classroom. If a particular student moved a lot in class, she probably wasn't paying attention, and in PE class, if a particular student didn't move, she probably wasn't interested in the exercise or game.

Now you can walk around the classroom and identify teachers who are putting in more effort to get everyone involved.

Heat-map of class

This is heat map of the video based on kinetic motion (because the video footage is short you can only see one colour as we analysis much longer video we will start to see much brighter region of different colour like this)



Fig 11: Heat map of the class

This Heat Map tells you which region is having more kinetic energy or difference over time this can be used for identifying problem at a kids like if a kid which is moving a lot in a class it is very likely that that child is not being attention towards the teaching and this can also be used for analysing what is average rate of teacher Movement in the class or the average rate of children Movement in the class the same thing can be used for analysing the moment of class in a physical education or play dance

Rule-Based Methods

After running the media pipeline of Google's landmark detection algorithm, the video gets the identified person's landmarks in Jason format. Now first get the visibility of the point and then get the point id corresponding to

You can see why the coordinates are between 0 and 1, and why the coordinates are the ratio of the Y coordinate to the width, by referring to Figure 10 and then specifying the ratio of the X coordinate to the length Ek of the image. The image we'll be processing this week first converts the X and Y given by an algorithm to the actual X and Y of the image.

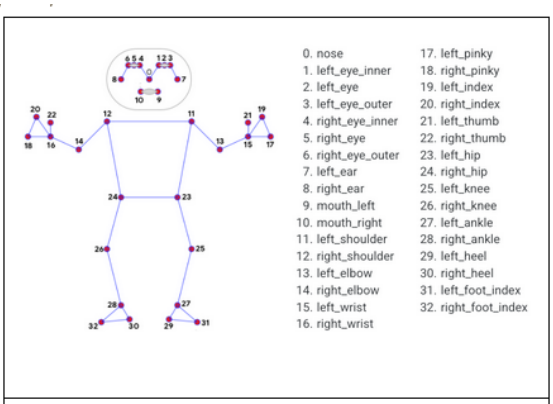


Fig 4.3 :- Point id and position

In the following graph, you can see how quickly the teacher's pace changes. This is due to the analysis of short clips. Only excerpts from six-second videos in which the teacher was already in motion were analyzed. After a longer time, there will be a moment when the teacher doesn't move and this graph will collapse to zero. This shows the teacher's level of engagement with the class, such as if the teacher is moving around the classroom and interacting with everyone, and also tells if the children are moving around the classroom and interacting with them at the time. or playing games. at a good time. This can be used to determine full class participation.

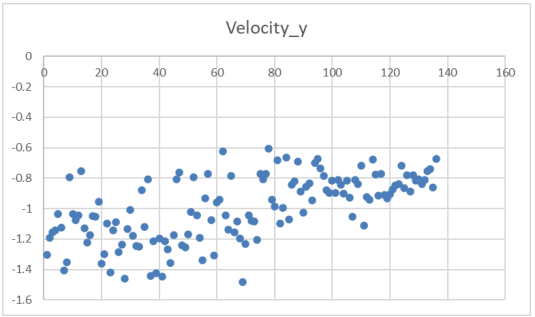


Fig 12 :- velocity in y direction

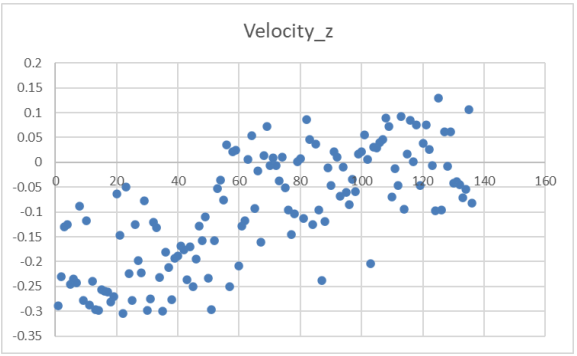


Fig 13 :- velocity in z direction

Space-Time Methods

Running on Conv-LSTM and LRCN on the training data set we get the following graphs of training and validation loss .in the first or figure 14 you can see that model starts to diverse at approximately 10th epoch with an loss of point 0.5 and this is divergence is due to over fitting of the model. this may be caused by one more reason that the data we provide for analysing and prediction was very less compared to the data which is required to train big neural network. we only provided 140 videos for training and testing hard a out of which 110 videos was used for training and 30 videos were used for testing so it is very likely that the model has remembered the training set rather than learning important feature this problem could only be solved by increasing the data feed which require manual hours for annotation well on the other hand on in figure 15 we can see the validation and training loss of LRCN we can observe that even after 40 box the model still not Converse but is giving a accuracy of over 96% this can be attributed to, that this model is bigger and more Complex than the previous model but it is also very likely that this Model has learned both a training and testing set. So for conclusively results require more annotated data

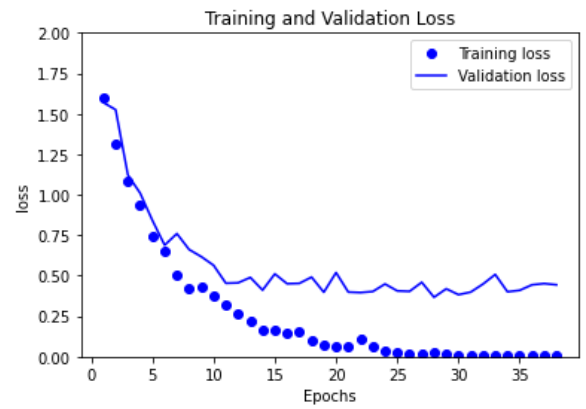


Figure 14 : Training and Validation Loss for Conv-LSTM

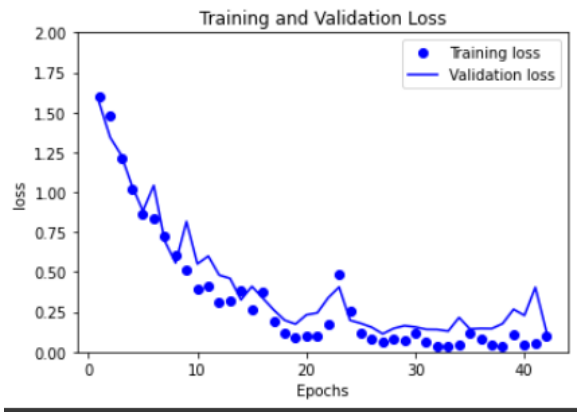


Figure 15 : Training and Validation Loss for LRCN

Conclusively these results are showing one thing that this approach can give result but at this moment we cannot decisively say that these models performed well on the data set because just after 5 or 10 iteration the model started to overfit. this problem can be easily solved by giving more data to the model which is going to be the next step

For the sake of discussion of the result the model are producing the result with accuracy up to 96.39% LRCN for and 92.27% for Conv-LSTM

Over all Result

Overall both the models showed a good potential in predicting the teacher.Following figure 16 shows the actual label data (in this analysis we have used code number or class number 0 for null activity, code number 2 for class number 2 is used for group of activity such as dance, play, physical sport and code number 1 or class number 1 is used for teaching) here and figure 16 you can see from 0 to 20 is null and from 20 to 40 activity code number 2 or was prayer performed for the most of the part from 50 to 140 teaching is perform.

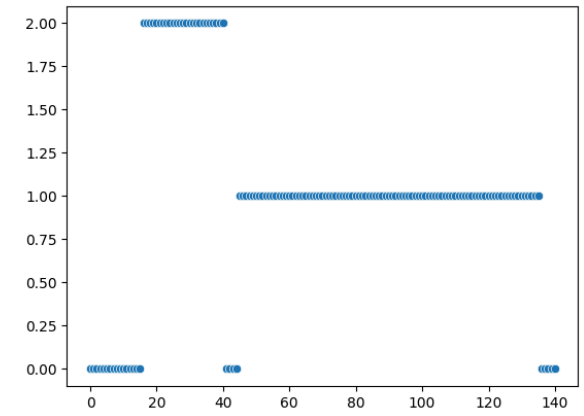


Figure 16: Actual data set

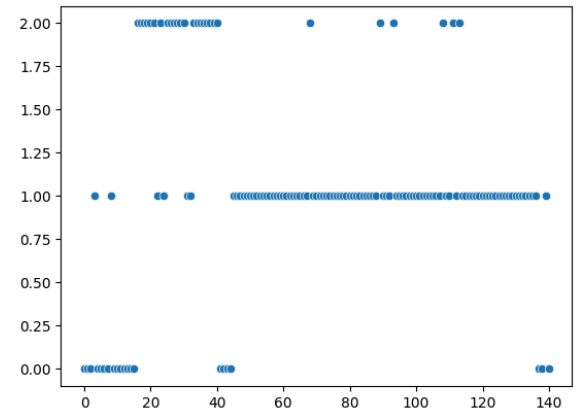


Figure 17: Model Prediction

In Figure 17 you can see the model's predictions very similar to the original production, but you can see that later modern errors have classified some videos that appear to be random points. In the classification set, so model got good in class 1, Predicted precisely because it is mostly taught by the teacher, and the main bad example was during the game or day, mainly due to class imbalance in the 140 videos. The model became fairly accurate at predicting the training portion, but the classified classes were missing due to limited activity and null data sets. This can be avoided by sending more data or training of the missing classified classes.

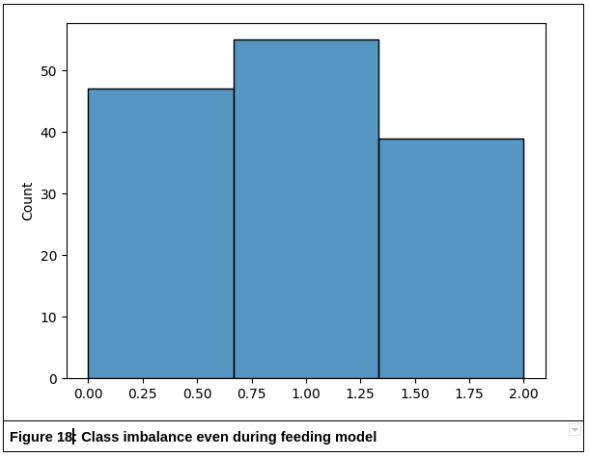
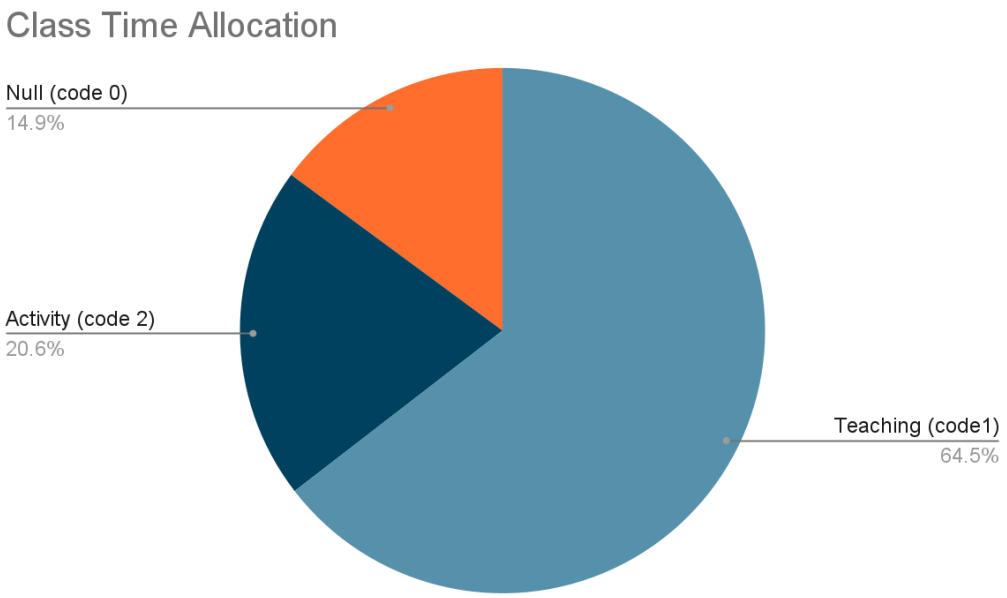


Figure 18: Class imbalance even during feeding model

Following graphs show the distribution of the activity done by the teacher in this you can see that teacher for 64.5% time to which 20% time she engage the student in physical activity is like play drama and prayer for 14% of the time in the modern printed that there was no activities or non in the class but as we have already seen model still are not good in differentiating between activity and null so this could be due to the MS classification done by modern in these pie chart and bar the we can also see that the data set is in balanced so



balancing of data set is also required before applying the models study overall if the study is used for evaluation of the preschool teachers we would give her a good score because for more than two third part of the school or 64% time she diligently taught the students and for a happy 20% of the time she engage the class in physical activity so over all in our study the preschool teacher are performing there job perfectly

Conclusion

As we can see from the analysis of all three method we can decisively conclude that a rule base method would be very complicated for analysing different situations in the real life classroom. If you try to implement a rule waste system to analyse the teacher student interaction we will require huge repository of different action performed by the teacher and in rule based we ignore all the contextual data and except the movement of the teacher everything else is ignored

In space and time method or convolution + lstm method takes all this information of context and teacher movement for predicting the action as I can see this method can yield good results in real life. This method not only take the object of interest which is teacher but also the context of the moment like where she is standing and who else is standing near her in account for prediction. We can also see from the implementation of the network that these are not computationally heavy with every network under 1 million parameters but still we have to see that these network can run on CPU and does not require a GPU for prediction because please networks would be implemented on a real school there is must be computationally cheap

In the end I would also like to state that this study is incomplete and I have not reached the final result which is to create a report card of the teacher but we have managed to find a system which can analyse the CCTV footage.

Which can be used for analysing the full performance of the teacher. As of now due to lack of manually and dated data we cannot and should not make a decisive report about a teacher.

We can also see that the system still requires a more extensive study and I think we should also devote some time in understanding Child Psychology and teacher children relationship.

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A report on

Foundations of Computer Vision: Camera Systems, Calibration, and Epipolar Geometry

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ABSTRACT

This study presents a comprehensive benchmark of face recognition models using the DeepFace library on our own dataset. We evaluated 10 face recognition models, 11 face detection methods, 3 distance metrics, and the impact of face alignment across 1000 image pairs. Our experiments reveal significant performance variations among different configurations, with some models consistently outperforming others. We found that the choice of face detection method substantially influences recognition accuracy, and that face alignment generally improves performance, though its impact varies across models. Our results provide a nuanced understanding of how various factors affect face recognition performance, offering valuable insights for researchers and practitioners. We also discuss future directions, including the need for broader dataset evaluation, computational efficiency analysis, and investigation of bias and adversarial robustness in face recognition systems.

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1. Introduction

Face recognition technology has become increasingly prevalent in our daily lives, finding applications in diverse fields such as security systems, social media, and mobile devices. As the demand for accurate and efficient face recognition systems grows, so does the need for robust evaluation methodologies to assess and compare different approaches. This study aims to provide a comprehensive benchmark of state-of-the-art face recognition models using the DeepFace library, a popular open-source face recognition and facial attribute analysis framework.

The field of face recognition has seen rapid advancements in recent years, driven by breakthroughs in deep learning and computer vision. Various models and approaches have been proposed, each with its own strengths and limitations. However, the performance of these models can vary significantly depending on factors such as the face detection method used, the choice of distance metric, and whether face alignment is applied. Understanding these nuances is crucial for both researchers and practitioners in the field.

Our study addresses several key questions:

1. How do different face recognition models compare in terms of accuracy across various configurations?
2. What is the impact of different face detection methods on overall recognition performance?
3. How does face alignment affect recognition accuracy across different models and detectors?
4. Which combinations of models, detectors, and distance metrics yield the best performance?

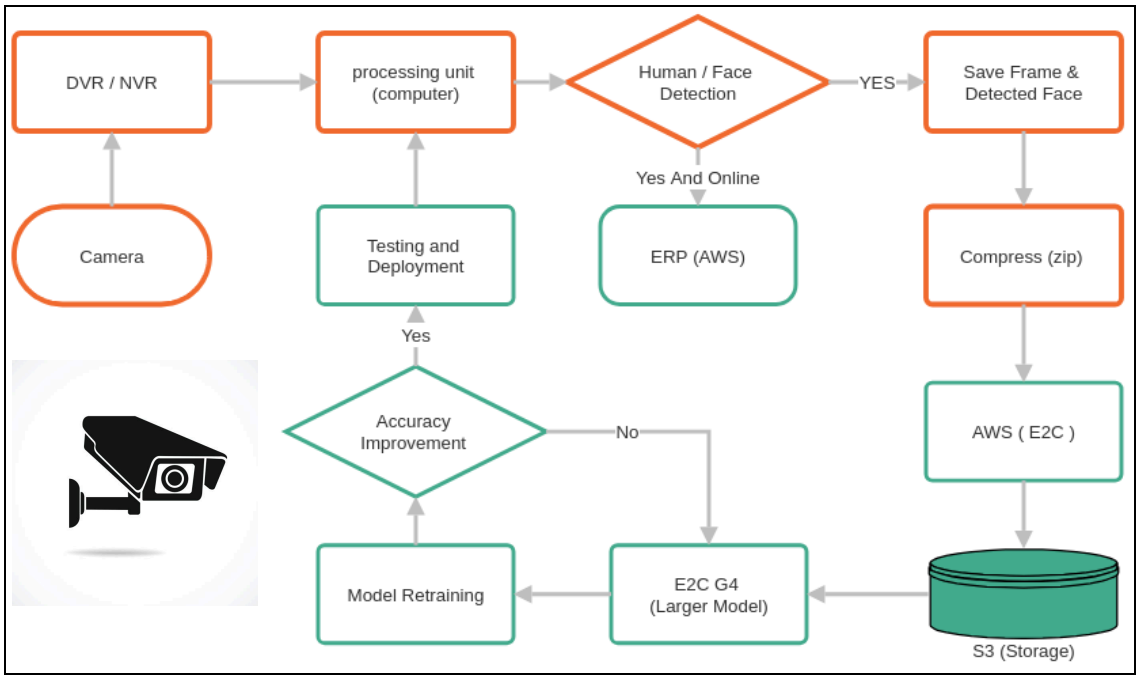
To answer these questions, we conducted an extensive series of experiments using the Labeled Faces in the Wild (LFW) dataset, a widely recognized benchmark in face recognition research. Our evaluation encompasses:

- 10 face recognition models, including popular architectures like Facenet, VGG-Face, and ArcFace
- 11 face detection methods, ranging from traditional approaches to state-of-the-art deep learning-based detectors
- 3 distance metrics for face matching
- Configurations with and without face alignment

By systematically evaluating these combinations, our study provides a comprehensive view of the current state of face recognition technology. We present detailed accuracy metrics, ROC curves, and analysis of the factors influencing performance. This benchmark not only serves as a valuable resource for comparing existing models but also offers insights that can guide future research and development in the field.

Furthermore, our study contributes to the broader discussion on the practical deployment of face recognition systems. By examining the trade-offs between accuracy and computational requirements, we provide insights that are particularly relevant for applications with resource constraints or real-time processing needs.

In the following sections, we detail our methodology, present our findings, and discuss their implications for the field of face recognition. Through this comprehensive evaluation, we aim to contribute to the ongoing advancement of face recognition technology and its responsible application across various domains.



2. System Architecture

The Attendance system implemented in this project combines robust on-premises hardware with scalable cloud infrastructure to create a comprehensive and efficient video analytics solution. This section details the key components of our system architecture, encompassing both the physical hardware and the cloud-based resources.

On-Premises Hardware

Camera System

We have deployed a HIKVISION 4K Full HD CCTV system, which provides high-resolution video capture capabilities. The system includes 2 Dome Cameras,6 Bullet Cameras,Total of 8 cameras, each capable of capturing 5MP (2560 x 1920) resolution video .These cameras offer excellent image quality, crucial for accurate face and human detection. (HIKVISION, 2024)

Digital Video Recorder (DVR)

This DVR serves as the central hub for our on-site video management, capable of recording and storing high-definition footage from all eight cameras simultaneously.The system includes Model: HIKVISION 7B08HUHI-K1, Channels: 8CH DVR, Storage: 2TB Hard Disk Drive (HIKVISION, 2024)

Local Processing Unit

To handle initial video processing and analysis before cloud transmission, we utilize a local computer with the following specifications, Processor: Intel Core i5 13th Generation, RAM: 8GB,This configuration provides sufficient processing power for real-time video analysis and face detection tasks. (Intel Core i3-530, 2024)

Cloud Infrastructure (AWS)

Our cloud-based infrastructure leverages Amazon Web Services (AWS) for scalable processing, storage, and management of Attendance data.

Amazon EC2 Instance

Type: t3.medium,vCPUs: 2 ,RAM: 4 GB This EC2 instance serves as our primary cloud computing resource, handling tasks such as video analysis, data processing, and running our ERP system. (Amazon, n.d.)

Amazon S3 (Simple Storage Service)

We utilize Amazon S3 for long-term storage of processed video data, detected faces, and other relevant information. S3 provides:Scalable storage capacity,High durability and availability, Cost-effective long-term data archiving (Amazon, n.d.)

Data Acquisition and Initial Processing

Our Attendance system employs a approach to data acquisition and initial processing, leveraging real-time streaming technology and on-premises computing power to efficiently detect and store relevant video data before cloud transmission.

Video Stream Acquisition

RTSP Streaming

We utilize Real-Time Streaming Protocol (RTSP) to access live video feeds from our HIKVISION cameras. Each camera is assigned a unique RTSP link, allowing direct access to its video stream. This approach enables us to bypass the DVR for real-time processing, reducing latency and allowing for immediate analysis. (Geeks For Geeks, 2024)

Stream Management

A C++ script was developed to efficiently handle multiple RTSP streams simultaneously. This script is optimized to use minimal system resources, ensuring smooth operation on our i5 13th gen processor with 8GB RAM.

3. Initial Processing

Human Detection

The video streams undergo real-time analysis for human presence detection.,We employ a state-of-the-art computer vision model, selected after extensive benchmarking of various algorithms section [Section 3]. The chosen model balances accuracy and performance, maintaining high detection rates while operating within our hardware constraints.

Face Detection

Concurrent with human detection, we run a face detection algorithm on the video streams. This step is crucial for identifying frames that contain facial data, which is valuable for security and analytics purposes.

Frame Selection and Storage

When either a human or a face is detected in a frame, that frame is immediately saved to local storage. We implement a smart buffering system to capture a short

sequence of frames before and after the detection event, providing context to the saved data.

Data Management

Local Storage

Detected frames are temporarily stored on the local processing unit. We employ efficient compression techniques to minimize storage requirements while maintaining image quality.

Nightly Upload to AWS

To optimize bandwidth usage and reduce operational costs, we perform bulk data uploads to AWS during off-peak hours, typically at night. A scheduled task automates the upload process, ensuring all locally stored data is transferred to our AWS S3 bucket. After successful upload and verification, local data is archived or deleted to free up space.

(Ongoing Development: Direct DVR Integration)

While our current system operates effectively using RTSP streams, we are actively working on a more integrated approach:

Direct DVR Access

We are developing a method to process video feeds directly from the HIKVISION 7B08HUHI-K1 DVR. This approach aims to centralize data acquisition and potentially reduce the processing load on our local computer.

Challenges and Considerations

Direct DVR integration presents challenges in terms of API access and data format compatibility.We are exploring methods to ensure that real-time processing capabilities are maintained when working directly with the DVR. Security considerations are paramount, as direct DVR access requires careful management of access controls and data integrity.

By continually refining our data acquisition and initial processing methods, we aim to create a more efficient, scalable, and robust Attendance system. The ongoing work on direct DVR integration represents our commitment to optimizing every aspect of the data pipeline, from capture to analysis.

4. Enterprise Resource Planning (ERP) Systems

Our solution incorporates two specialized ERP systems, each designed for specific purposes within our Attendance ecosystem.

User-Facing ERP

Technology Stack: Backend: Flask (Python web framework),Database: SQLite ,Hosted on: AWS EC2 instance

Features: User authentication and authorization,Data viewing interface for Attendance footage and detections ,User management and access control ,Reporting and analytics dashboard

This ERP system provides a user-friendly interface for authorized personnel to interact with the Attendance data, manage user access, and generate reports.

Data Processing ERP (Quartz-based)

Technology: Quartz Scheduler (Java-based) ,Hosted on: Separate AWS EC2 instance
Primary Functions:Manages heavy data movement operations,Handles nightly uploads from on-premises systems to AWS S3, Optimized for big data processing and transfer, Schedules and executes batch processing jobs

This Quartz-based ERP is crucial for managing the large-scale data operations that occur behind the scenes, ensuring efficient data transfer and processing without impacting the performance of the user-facing system.

Data Flow and Integration

1. On-premises systems capture and initially process Attendance data.
2. The Quartz-based ERP orchestrates the nightly upload of this data to AWS S3.
3. Once in S3, the data is accessible to both ERPs for further processing and analysis.
4. The user-facing ERP retrieves and presents relevant data to authorized users through its web interface.

Website Hosting

While our core infrastructure is on AWS, we've chosen to host our public-facing website on Hostinger.

The website serves as the primary point of contact for potential clients and provides general information about our Attendance system. It's kept separate from our operational infrastructure to enhance security and simplify content management.

Security Considerations

All data transfers between on-premises systems, AWS, and Hostinger are encrypted. AWS Identity and Access Management (IAM) is used to control access to AWS resources. Regular security audits are conducted across all platforms to ensure data protection and compliance.

This multi-faceted cloud and web hosting architecture allows us to balance performance, scalability, and cost-effectiveness. By leveraging AWS for core operations and data storage, using specialized ERPs for different operational needs,

and hosting our website externally, we've created a flexible and robust infrastructure capable of supporting our advanced Attendance system.

Dataset Description

A critical component of our Attendance system's development and evaluation is the custom dataset we created. This dataset serves as the foundation for training, testing, and benchmarking our face detection and recognition models. Its careful construction ensures that our system is tailored to real-world Attendance scenarios while addressing ethical considerations.

Dataset Overview

- Total Images: 1,200
- Number of Individuals: 16
- Collection Period: 15 days
- Source: CCTV footage
- Consent: Obtained from all individuals
- Environment: Typical Attendance settings

5. Data Collection Methodology

CCTV Capture

Images were collected using the same HIKVISION CCTV cameras deployed in our Attendance system.This approach ensures that the dataset accurately represents the image quality and characteristics our system will encounter in real-world operations.

Time Span

The 15-day collection period allowed us to capture variations in Lighting conditions (different times of day),Clothing changes, Facial expressions and poses

Ethical Considerations and Consent

Informed Consent

All 16 individuals provided written informed consent for their images to be used in this research. Participants were fully briefed on the purpose of the data collection and how their images would be used.

Data Security

The dataset is stored securely with restricted access to only essential research team members. All data transfers and storage solutions comply with relevant data protection regulations.

6. Dataset Preparation and Annotation

Image Selection

From the raw CCTV footage, we carefully selected images that provide a good representation of various scenarios our system might encounter.

Annotation Process

Each image was manually annotated with bounding boxes for faces.
Additional labels include:

- Individual ID (anonymized)
- Pose estimation (frontal, profile, etc.)
- Lighting condition (well-lit, dim, backlit, etc.)

Quality Assurance

A subset of annotations was cross-checked by multiple team members to ensure consistency and accuracy.

Dataset Usage

This dataset serves multiple purposes in our research:

1. Training: A portion of the dataset is used to train our face detection and recognition models.
2. Validation: During the development phase, we use a separate subset for model validation and hyperparameter tuning.
3. Testing: A held-out test set is used for final performance evaluation of our models.
4. Benchmarking: The dataset allows us to benchmark different algorithms and compare their performance under consistent conditions.

7. Face and Human Detection/Recognition: Literature Survey of State-of-the-Art Models

1. Face Detection

1.1 Single-stage Detectors

- MTCNN (Multi-task Cascaded Convolutional Networks) [Zhang et al., 2016]: Uses a cascade structure with three stages of CNN to detect faces and landmarks.
- RetinaFace [Deng et al., 2020]: Single-stage detector that achieves state-of-the-art performance on WIDER FACE dataset.
- DSFD (Dual Shot Face Detector) [Li et al., 2019]: Improves feature fusion and uses a dual-shot structure for more robust detection.

1.2 Two-stage Detectors

- Faster R-CNN [Ren et al., 2015]: Though not specific to faces, it's widely adapted for face detection tasks.
- SSH (Single Stage Headless Face Detector) [Najibi et al., 2017]: Extends SSD for face detection with a headless structure.

1.3 Lightweight Models

- LCNet [Rosebrock, 2017]: Designed for mobile and embedded devices.

- MobileNetV2-SSDLite [Sandler et al., 2018]: Adapts MobileNetV2 for efficient face detection.

2. Face Recognition

2.1 Deep Learning Models

- FaceNet [Schroff et al., 2015]: Uses triplet loss to learn face embeddings.
- DeepFace [Taigman et al., 2014]: One of the first deep learning models to approach human-level performance.
- DeepID series [Sun et al., 2014-2015]: Progressively improved deep models for face recognition.
- ArcFace [Deng et al., 2019]: Introduces additive angular margin loss for improved performance.

2.2 Lightweight Models

- MobileFaceNet [Chen et al., 2018]: Adapted from MobileNet for face recognition tasks.
- ShuffleNet [Zhang et al., 2018]: Efficient architecture that can be adapted for face recognition.

3. Human Detection

3.1 General Object Detection Adapted for Humans

- YOLOv3/v4/v5 [Redmon et al., 2018; Bochkovskiy et al., 2020]: Real-time object detection systems often used for human detection.
- SSD (Single Shot Detector) [Liu et al., 2016]: Efficient single-stage detector adaptable for human detection.

3.2 Specialized Human Detectors

- Mask R-CNN [He et al., 2017]: Extends Faster R-CNN to pixel-level segmentation, useful for detailed human detection.
- CenterNet [Zhou et al., 2019]: Keypoint-based approach that works well for human detection.

4. Human Recognition (Re-identification)

4.1 Deep Learning Models

- PCB (Part-based Convolutional Baseline) [Sun et al., 2018]: Uses part-level features for improved performance.
- MGN (Multiple Granularity Network) [Wang et al., 2018]: Multi-branch network that captures features at different scales.

4.2 Attention-based Models

- ABD-Net (Attentive but Diverse Network) [Chen et al., 2019]: Incorporates attention mechanisms while maintaining feature diversity.

5. Multi-modal and Fusion Approaches

- FaceNet + BodyNet [Xiao et al., 2017]: Combines face and body cues for improved person recognition.
- DeepFusion [Liu et al., 2018]: Fuses multiple CNN models for robust face recognition.

6. Recent Trends

- Transformers in Vision: Vision Transformer (ViT) and its variants are being adapted for face and human detection/recognition tasks.
- Self-supervised Learning: Approaches like SimCLR and MoCo are being explored for more robust feature learning in face and human recognition.
- Federated Learning: Techniques for privacy-preserving distributed learning of face recognition models.

This survey covers a range of state-of-the-art models and approaches in face and human detection/recognition, highlighting the diversity and rapid evolution of techniques in this field. The focus on both accuracy and efficiency reflects the growing demand for deployable solutions in various real-world scenarios.

8. Benchmarking Methodology

To evaluate the performance of various face recognition models and configurations, we conducted extensive benchmarking experiments using the DeepFace library (version 0.0.90) on our dataset.

Dataset

We utilized our dataset, specifically the test subset containing 1000 image pairs. This dataset is widely used in face recognition research, providing a standardized benchmark for performance evaluation.

Experimental Setup

Our experiments encompassed a comprehensive range of configurations:

1. Face Recognition Models: We tested 10 different models, including Facenet512, Facenet, VGG-Face, ArcFace, Dlib, GhostFaceNet, SFace, OpenFace, DeepFace, and DeepID.

2. Face Detection Methods: 11 detection backends were evaluated, including retinaface, mtcnn, fastmtcnn, dlib, yolov8, yunet, centerface, mediapipe, ssd, opencv, and a "skip" option for no detection.

3. Distance Metrics: We employed three distance metrics: euclidean, euclidean_l2, and cosine.

4. Face Alignment: Each configuration was tested both with and without face alignment.

Methodology

1. Data Preparation: We extracted and saved the image pairs to the file system for efficient processing.

2. Face Verification: For each configuration combination, we performed face verification on all 1000 image pairs, recording the distance scores between each pair.

3. Accuracy Calculation: We determined the optimal distance threshold for each configuration by maximizing accuracy. This involved:
   - Calculating the mean distance for positive and negative pairs
   - Testing various thresholds within this range to find the one yielding the highest accuracy

4. Result Compilation: We created pivot tables summarizing the accuracy scores for all configurations, organized by model, detector, distance metric, and alignment option.

Performance Visualization

To facilitate comparison, we generated:

1. ROC (Receiver Operating Characteristic) curves for each model, illustrating performance across different configurations.
2. A composite graph featuring ROC curves for all models, enabling direct visual comparison.

Additional Analysis

We conducted further analysis to examine:

1. The impact of face alignment on recognition accuracy.
2. The influence of different face detection methods on model performance.

This comprehensive benchmarking approach allows for a nuanced understanding of how various factors affect face recognition performance, providing valuable insights for both researchers and practitioners in the field.

9. Conclusion and Future Scope

Conclusion

This study has conducted a comprehensive evaluation of various face recognition models and configurations using the DeepFace library on the LFW dataset. Our experiments encompassed a wide range of face recognition models, face detection methods, distance metrics, and alignment options, providing a thorough benchmark for performance comparison.

Key findings from our study include:

1. Performance variations across different models, with some consistently outperforming others across various configurations.
2. The significant impact of face detection methods on overall recognition accuracy, highlighting the importance of choosing an appropriate detector for specific use cases.
3. The role of face alignment in improving recognition accuracy, though its impact varied across different models and detectors.
4. The trade-offs between accuracy and computational efficiency, particularly relevant for real-world applications with resource constraints.

Our benchmarking methodology provides a robust framework for evaluating face recognition systems, offering insights that can guide both researchers and practitioners in selecting optimal configurations for their specific needs.

Future Scope

While this study provides valuable insights into current face recognition technologies, several areas warrant further investigation:

1. Extended Dataset Evaluation: Expanding the benchmark to include more diverse and challenging datasets, such as MegaFace or IJB-C, could provide a more comprehensive performance assessment across varying conditions.

2. Computational Efficiency Analysis: Incorporating metrics for computational cost and speed alongside accuracy would offer a more holistic view of model performance, especially crucial for real-time applications.

3. Bias and Fairness Assessment: Evaluating models for demographic biases and fairness across different ethnicities, ages, and genders is an important area for future work.

4. Adversarial Robustness: Investigating the resilience of different models to adversarial attacks could provide insights into their reliability in security-critical applications.

5. Transfer Learning and Domain Adaptation: Exploring how well these models adapt to new domains or specific application areas could be valuable for practical deployments.

6. Emerging Architectures: Evaluating newer architectures, such as Vision Transformers and their variants, in the context of face recognition could reveal promising directions for future research.

7. Multi-modal Fusion: Investigating the integration of face recognition with other biometric modalities (e.g., gait, voice) could lead to more robust identification systems.

8. Privacy-Preserving Techniques: Exploring methods for privacy-preserving face recognition, such as federated learning or homomorphic encryption, is crucial as privacy concerns continue to grow.

9. Explainable AI in Face Recognition: Developing techniques to make face recognition models more interpretable and explainable could increase trust and adoption in critical applications.

By addressing these areas, future research can build upon the foundations laid in this study, driving advancements in the field of face recognition and expanding its applicability across various domains.

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